

ActInf Textbook Group ~ Cohort 6 ~ Session 21 (Applying Active Inference)

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all right September 9th 24 and Andrew go

for

it um hi everyone so uh as we just

briefly I know some is coming in uh as

we just briefly discussed uh since today

is another session where we are not

directly talking about any particular

the chap chapter of the textbook but

rather um just talking about applying

active inference more generally um we

agreed that I would spend just a little

bit of time here uh giving sort of a

quick rundown of how to construct an

active inference agent in this case

it'll be

specifically uh a PDP agent meaning

we're working with um well in this case

we're working with discrete time and

discret

observations um as opposed to continuous

time models which you'll see in chapter

8 of the

textbook um and I kind of just want to

focus on just from scratch like how do

we think about constructing an agent um

I won't be touching on how to construct

an environment um but but that's the

kind of thing that you know you would

you would bear in mind whenever kind of

coming up with an idea for a simulation

or experiment you want to run um and

actually on that note Daniel is it

possible for me to um share my screen by

chance yep go for it awesome thank

you so this was the um and if anyone

wants this link they can find me on
GitHub a p hea just like my last name on
the zoom uh and the repositories called
ic2 S2 active inference tutorial um this
is material that I presented back in
July at University of
Pennsylvania um I
basically uh gave a tutorial on AC
inference how it might be applied to
social sciences specifically in uh
constructing multi-agent
simulations
so um and is
this uh is this visible to everyone the
slides
here yeah go for it for you know five to
10 minutes however much you just want to
footnote and then let's see what people
want to continue with definitely sounds
good um so in this tutorial
along the way I you know we're
introduced active inference there's some
nice um explanations of different
concepts from the textbook Etc talking
about uh the basy brain hypothesis
General ideas how Bas rule kind of plays
out in active inference and eventually
we reach where uh we're going to use the
pmdp package um which has been developed
to make discrete time AC inference
agents in Python and um there are a lot
of updates being made to that package
right now if you were to like use a git
clone to download it directly there's a
lot going on in there but for the time
being I'm going to just be using the the
PIP install um Pi Pi
version uh it's a little bit more kind

of uh packaged up comes as more like a single unit so um whenever we construct active inference agents uh the starting step should be in my view starting step should be what are the hidden states that the agent has in their generative model like what are they trying to infer uh second thing is what are the observations that the agents will encounter in the world and then three what are the actions uh or controls or policies those three terms are sometimes used interchangeably they don't necessarily mean exactly the same thing um but for now we'll just go through with all of those we'll call them actions uh what actions can the agent commit uh so so we can view that kind of like like this like the this is the agent that we be constructing um all of these things play out it's called an action perception Loop um let me see if I can just okay hopefully that's visible enough to everyone um so so that we'll we'll start at the bottom of this action perception mod um first observations there are two kinds of observations that this agent will receive at any given time step um quick note that this entire action perception Loop plays out at every time step we're working in discrete units of time and so you know whether it's from second to Second minute to minute millisecond a millisecond however um all of the units of time are the

same uh here the agent receives one of two observations one of them is related to uh an improve Improvement if something improves or if it doesn't improve and I didn't include a third category but there's another one called unobserved just bear with me and we'll go into what that means soon um but that's one particular kind of observation otherwise known as an observation modality that the agent has slashes we can think about about this is like in the same way that we have you know sensory data that comes in uh you know like you have cold receptors on your on your uh in your scan on your arms so you can feel if it's cold or it's hot or otherwise similarly here the agent has some hypothetical receptor and we could call it visual or otherwise but the point is they they get one of these three observations Improvement no improvement or an observed um another observation modality they have is self or neighbor and I call this the attention modality uh at any given time step they're paying attention either to themselves or to their neighbor um part of why we're talking about self neighbor that kind of thing is that eventually this leads to a multi-agent simulation where where agents can can either uh kind of do their own thing or they can interact with other agents and so this attention modality means they they're actually observing like who am I paying attention to um so when they take in those

observations they have a single hidden State uh which is I've denoted s with superscript ATT this is their attention hidden State um an agent takes in observations and then using what's called a likelihood model which is uh uh in in in chapter 7 we refer to it as the a matrix remember we have the the a b c and d matrices uh this is the a matrix and so the a matrix is this it's the probability of that observation that they just received given a hidden State uh whenever the agent reaches this this right uh this Square on the right so to speak this stage of action perception cycle uh they invert their models so now they they actually try to figure out what is the hidden State given I just received this observation you know in the same way that you could receive a a hot or cold observation and so you infer oh it's it's cold here um here the agent is for example they might observe themselves and then infer oh I'm paying attention to myself um and vice versa for neighbor here this is an incredibly uh despite the material despite kind of the complexity of active inference Theory this agent is actually very simple they can receive one of two observations every time step they have a single hidden state that they're trying to infer uh self or other but what that does is that it makes a link between if they observe an improvement or no improvement uh with who they're paying attention to and whenever you combine that with the idea of learning their

likelihood model which is what this
Delta this triangle means like they're
learning their likelihood model um they
can learn that oh if I pay attention to
myself I see an improvement more than if
I pay attention to my neighbor uh right
so so so these kind of values get these
parameters get adjusted over time these
are the agent's beliefs changing based
on their learning from experience and
then after they make an inference they
then decide okay which action should I
take on this time step uh the names of
the actions are explore and exploit
which is um that doesn't quite have
anything to do with the the so-called
explore exploit dilemma that comes up in
the
textbook um you know that's technically
a different thing these are just
arbitrary names it explore here really
just means the agent is exploring
new Solutions this is a problemsolving
agent the idea is they can explore
meaning on their own pay attention to
themselves and look try and come up with
new solutions to a problem or they can
exploit meaning that they will steal the
answer of their neighbor right and if
they so so you can see how if they steal
the answer from their neighbor and they
see an improvement then they start to
learn like oh if I just keep taking my
neighbor's answer to this problem I keep
seeing improvements or it might be the
other way around I might see no
improvement start to associate like ex
the action

exploit receive the observation neighbor
and no improvement and then in their
likelihood model they'll learn like oh I
shouldn't steal from my neighbor that
least to you know know progress on me
trying to solve this problem now I've
already discussed these three kind of
squares these three not of of this graph
here um that's all you need for an agent
uh the only thing I've I've left out is
the environment and it's the environment
that you know for a a programmer you can
think like oh well if I set up an agent
who has observations and in state to
infer in actions that just gets hooked
up with the environment the environment
can be a list of you know if else roles
like if the agent does this then the
environment should send out this
um or it could be more of a stochastic
process you could use you know a random
number generator or
otherwise um so the agent will learn
over time so if anyone is uh interested
in this again feel free to to check out
the uh I'll actually put the link in the
chat whenever I'm done here I don't want
to take up too much time though but it's
just that's the general process of an
action perception Loop and then you know
the tutorial from there just kind of
starts to break it down even further
like once you so once you figure out
those hidden States like what is the
agent trying to infer what are the
observations that can receive what are
the actions that can commit The Next
Step would be okay how do we link those

three things together through your a b c
d and e um you know a links together
observations States be links together
transitions between states with what the
agent does it's like oh if I think X
right now and I do y then I believe that
c will happen next that's what kind of
like what a b Matrix says C Matrix prior
over uh observations which is like your
preferences for what you want to see the
agent will kind of try to realize these
or seek out these observations by way of
free energy minimization uh D is your
initial beliefs about States you know
what did you think before um that and
then and then through uh inference we'll
we'll end up with a state uh belief a
posterior that might differ from D or it
might be exactly the same in which case
that means that the agent just
reconfirmed its own belief again it's
exactly the same and E just your priors
over actions so what are you most likely
to do regardless of anything uh that's
why they're also known as habits um I'll
I'll stop that there's there's plenty
more to go into but those are all the
core
barebone aspects of what you need for an
agent and the rest is just you know I
pretty quickly start jumping into code
from there on how to do it so um yep
that's it for now
so oh sorry can I have the link to that
I'd love the link to that in absolutely
yeah worries I mean that's right my
wheelhouse I love this stuff bu and I I
just spend all my days doing this kind

of things so I'm it's nice to see somebody else like building out like the step-by-step process of like how do we like make this work kind of and then you get to like just throw it all you know that way other people can do it too yeah cool yeah yeah exactly the point yeah I like for me personally it's like it's only uh I'd already been involved with the institute for some time and then realized like there is no clearcut way of doing this um I just dropped the link in the chat or sorry uh it's not the specific um but in any case yeah so so what I did was that I just started getting the documentation for PDP and I did a lot of like just going through the repository to make sure I'm getting things right um there's a lot of complexity because they tried to build imdp off of SP m in mat lab which is what Frist and other folks developed and it's just been in development for years so they're trying to recreate this massive project and uh I just think like well if you want to get more people using it let's make some more kind of from scratch uh tutorials on how to go about this so that's that's what I did but cool I'm glad glad you find it interesting oh yeah I mean I had to come at it from a complete different uh uh hold on I get back in this thing here I lost all my stuff um yeah the uh I came at it from a completely different angle and but came to the same conclusions and it's I had no idea that somebody was

doing that it's so much
fun there we go there's everybody yeah
there there's a lot to add I think one
of the biggest pieces then anyone who
wants to ask go for it is like what is
the question about an active inference
generative model what's a question that
staples into A B C or D or a line of
code or a specific node or Edge on a
graph what's a question about the world
so it's like what's a question about the
subway map of the city what is a
question about the territory of the city
that can get back and forth and we know
all about and talk all about these kinds
of ambiguities and and dual reference
and things like that but at the same
time it is a different type of question
to say what is the dimensionality of the
a matrix
in comparison with what is the
dimensionality of agent sense making or
something like that and to know when one
is clearly on one side of the other or
when you're not sure is some of the
fastest ways to start separating kinds
of questions according to is this
something that's just going to be
addressed in a line of code or a math
equation or is this something that's
more like a real world line of inquiry
that cannot be answered in any string of
words that somebody says
I feel like if you can't answer it then
it you
know it's like that I mean there's a lot
to be said for not being able to answer
something right like um but uh I

definitely have a lot of questions um
I'm excited to read the material so I
can like start asking those in a little
a little better way I'm trying to think
right

now yeah Zach go for it oh I was just
gonna ask if you guys think
are there any questions that you
fundamentally can't answer or questions
that we just don't have the ability to
answer this might be a little bit more
philosophical but I feel like Daniel you
were kind of leading to potentially the
question of what is it like to be a bat
and maybe ideas that kind of into that
realm talking more about something about
the agent and its relationship to the
environment instead of not instead of
anything I'm sorry that's the end of the
sentence I just want to say really
quickly that uh this reminds me of
um a really crucial point they make in
the textbook uh I I think it's in
chapter nine but there's this uh early
on in the chapter it it must be chapter
nine but I might be mistaking that but
it has to do with analysis and the whole
point is they have this figure that
shows like okay on the
innermost uh kind of square of the
figure we see the typical like graphical
model that they you know A variation of
that they keep showing through the
textbook but then on the outside like
there's a square around it denoting like
okay it's really making the point that
this is what we the scientist or
experimenter think is the model of the

agent yeah thanks Daniel that we're trying to develop right it's just like really trying to make the point that like we ourselves don't firmly know necessarily what the inside of the mo like a rat in a teammates like I'm trying to model it I have to be aware that like here's the level that is what I think is the rat's perception of what's going on and how it learns and so on but you know so it's like the objective model is just like outer like this is what I'm constructing like you're in the picture too is how I yeah great point and and for examp ex Le the experimental actions of the experimenter are just it's just taken as an observation train we're not modeling the policy that led to the experiment being designed okay but you could include that so you can continue extrapolating um but it it also makes clear where where the the buck stops but about the questions it can't be answered I mean big question right but there's a sense in which behaviorally every question is always answered if with acknowledgement a postulational head not dare I say or silence um or or or some other kind of like nons respon so you know can be answered in Trivial sense yeah everything already is can be answered in more like a Justified true belief semantic validity sense so here's a few pieces from the ontology project built up over the years but I mean just a snapshot if someone wants to dive in further and add more it's the greatest

types of contributions that can help at this on logical level here's like many many many of the phrases and writings in the textbook group are like how is this term related to this um when they said this did they mean that does this mean this what does xers think does X why would X person want to believe that this was an explanation of that I mean whether or not you go into it questions have these templated forms so clarifying like what is active inference what surprise um going into those and knowing that even with the awareness of the question like what is surprise or how are they using surprise here those are already categories of questions that you can go in with a structural map towards and another fun um Direction with the with the philosophical side with like Oli and and Paulo Maria and others in the ontology was like starting with the Linguistics of the you know how could we ask and proceed on a question without like knowing what was written from a symbolic perspective so it's like okay then what is the the the lexical semantics like did they say something in the paper here we here we Define a for here we use surprise to mean this or here we operationalize surprise as so it's like they're narrowing down or leaving implicit what they mean by surprise but if they said we by surprise we mean one and then a person constructs an argument where it's predicated on surprise being number two then it's off base then getting into their local

weaving of the relational semantics so
not just how they Define the terms but
it's like okay given that they said
blanket states are sense and action you
know then did it make sense with this
other usage of
set that's in the internal reading
internal
consistency view from inside like was
there an
apparent issue without even going a view
outside of it then getting into views
from the outside oh I think that they
didn't do this or not I don't think
enough attention was paid to this they
should have done this I think it was
cool that they did this
um what would an embodied person think
about this like all manner of then
externalized but it's kind of just like
a combination of like a close and
deliberate and charitable structured
reading in that with reference to the AC
in ontology but it could just be any
ontology system at that point you give
the internal
reading and understand when questions
and methods are being used that are
related to internal
reading and then when is it like an
external reading from like a different
perspective or a later
time what do you think Zach or what what
kinds of questions does that does
that answer or not I mean that makes it
seem I really like the external idea of
looking at it for many many different
lenses because then we're naturally

going to find ways to maybe collapse
some language and find other things that
might have been equal in places that we
weren't expecting to look I think just
translating into a different language
can have a ton an amazing amount of
positive impact and it just makes
collaboration a really cool thing but at
the same time

I I've always been attracted to active
inference because it has this uh physics
backing it all it all comes from the
principle of least action and everything
about that so in my head there should
always be that physical explanation for
all things or at least a mathematically
based explanation for all things which
really speaks to me

personally man if I can jump in here I
really appreciate that because it really
speaks to me personally too like being
able to pull things out of the out of
the the discourse and put them into
theoretical Frameworks mathematical
Frameworks gives us it's like that low
and high road that active inference you
know it gives us a path to creating
tangible Solutions around these
things and I I think it's like it's kind
of esoteric almost in a way I mean the
you know Daniel uh I think and I were
talking last week about how for instance
it's this is so valuable for people who
love all manner of Sciences because
they're like uh it it intersects
everything philosophy and on and on and
on you guys know um and
um it does I feel like it necessarily

requires like atic perspective an
external perspective to um and objective
perspective to like write about these
things meaningfully you have to touch
base on everything and you guys should
see what a lot of this stuff does when
you uh put it into different uh
indigenous languages
pretty yeah you're absolutely right the
formalism
100% there so many ways to go with but
like modal logic like the shirt might be
blue or or conditional Logics or
paraconsistent all these other kinds of
of alternative logical structures where
things are not like either one way or
another outside of an evaluation Etc
it's like if someone says well ramstead
claimed X and I think he's saying it's
like that cannot be an incorrect second
half of the
sentence unless it were were deceptive
but if it were deceptive like the person
was like I think that ramed means one
but I'm going to verbalize that I think
he meant to that still would be their
veritical behavioral
representation so it would truly be what
they had said at that time and they can
have some other side information like um
but I said that because of this or I
would have said that so it's like but
they can't be
wrong so it when the space of ideas is
so vast there's so many PES more than
we'll ever read Plus+ plus generative
intelligence generating the
dissertations we don't have yet on

demands and and
more expressions are not opposed to each
other even when they contain what
appears to be like the clearest possible
like this person thinks that the blanket
means X this person says it's not X and
their their epistemic frames are all
aligned it's like even in that kind of
idealized scenario all the quantum
reference frames were aligned we had
gotten down to it and then two people
differed and one said it was up and one
said down still that situation could be
analyzed as a cognitive environment
where they both truly had those
perspectives and then a third Observer
could take another take on it and it
would also only add to the
situation to piggy back off of that I
think that's exactly what we saw in
geometry in the early 1900s with the
airong program where essentially there
were a bunch of geometries there was a
parabolic geometry which I don't think
that's how you pronounce that now that
I've said it uh there's uh this
algebraic geometry all of these
different geometries that all seem to
talk about very different things they
all went in and were able to describe
certain theorems but they all took all
of these amazing paths around and none
of them seemed to intersect until the ER
longan program which was able which was
of course the mathematician erir longan
or was it the place airong in forgive me
it's been a minute since I've read it it
was the birth of gauge Theory they were

able to take all of these seemingly
disparate ideas and find a relational
way to talk about all of them which gave
us gauge Theory today which actually
we're seeing in a lot of places it led
to Something in machine learning as well
called the 5gs which I won't get into
detail but it is the main method that we
have for trying to understand learning
itself as a method
to compress as a method of equating
intelligence with compression into a
different space which is just amazing it
really does remind me of kind of the
method of scientific revolutions if you
guys are into Dusty old books from the
60s I love Dusty old
books man ER

Paul the gauge Theory like also if we
think about the recent development of
actin so it's like 2022 textbook it's
like oh yeah wait a minute we're in the
textbook group 2022 textbook it's like
step by step by Smith at all also
leveled up with some more context and
it's published as a book it's a
state-based
approach it summarizes and crystallizes
the state-based approach to active
inference that's why the discussion is
like discrete State spaces continuous
State spaces for variables and for time
itself as a variable in the model then
especially from 2019 not so much
reflected only alluded to in the 2022
textbook coming on much more in papers
and in sanj nam
josi's future volume two book but the

basics and mechanics live stream 49 all of those kinds of things just opposing the state-based formalisms for active inference textbook discreet continuous Etc with path-based formalisms then ramstead and Dalton pointed towards what they call G Theory and highlighted like different deep dualities and some of their implications as well as the gauge theoretic aspects of what was being discussed like overall so it's a great tie in with with Gauge Theory there's so much to to to um build and and clarify around that um so it's interesting that you brought it in but yeah really interesting I'm sorry but uh I think is that the second time you brought up ramstead could you please say a little about him I'm simply not familiar I'm just like I'm just like using him as as like um you know he's he's an active discourse ant and and a great contributor from the pre 2010 PhD student philosophical and his his deep knowledge there and then on through more the applied um and and ongoing um but using it like as sort of just he's out there making or you know friston says X it's just anyone's reported speech um yeah Federated inference yeah yeah that's yeah yeah yeah I was just digging into that oh yeah this was uh I was just sharing this because I'm personally still new to the notion of um paths here and uh I think I

think in this paper they replace preferences like like like agents prefer a particular path and to me that that seems to imply like they're not just preferring like particular observation a particular moment in time they're preferring like entire sequence um not sure if I'm mistaken there or not but I was really interested in in that not done reading it yet these are great questions also natural language terms do get overloaded you know with with um with Grievous unnecessary and sometimes requisite overloading and underloading and ambiguity so like a path but then we might say well but the path that they take through space over time one two three we're going to call that a path so then it's like but that's not this one and then you look at from again from an ontological analysis okay paths may or may not depend on action it'd be like wait a minute what how could the path taken in the world not depend on action because path is being used in multiple subname spaces so it be like having a python script where like a variable is defined within an Loop and so it's used the way it's used within a loop but then there's a separate Global context or there there's two functions with the same name so disambiguating the Name space is some of the most important work each of these can have like a unique um you know star code ID fractal hash so that the local semantics can be

described and and modulated or
falsified it's like here's what what we
here's where we came down on the
semantic Vector for how path was used
here does your vector on that usage of
path
differ and then can can explore like
many many more layers of of semantic
similarity and
difference but yeah I don't I don't know
if this is actually using
it um in the path integral based basian
mechanical sense because this is still a
state space based
model yeah see yeah it's I I was
surprised they didn't go with the word
State you
know yeah it's what was interesting is
uh when you did the the control F and it
highlighted all of them the last half
highlight there and paragraph the full
sentences the remaining tensors encode
prior beliefs about paths C and the
initial States D and I think that brings
up another interesting point of trying
to kind of parse through the terms uh in
that oftentimes like sometimes the sort
of hint of how someone might have been
to term is that they related to
something else that might be known from
an you know whether it be from an
entirely different context or not like
here they're using the same like
alphabetical terms uh to describe like
for a it's still a likelihood Matrix for
B it's still a transition Matrix but
then here C which is typically in in the
textbook in chapter 7 in that particular

context C refers to your prior over observations but here they're now prior beliefs about paths and meanwhile once again initial prior about initial States D Remains the Same just does a b do so that's sort of like you know it's like this took me off guard when I was reading this I'm like everything here makes sense to me just intuitively right off the bat I've written a tutorial like putting together certain kinds of BPS so uh but then swap out uh preferences or observations for the word paths and I'm not really really sure but um right and then it's like wait but is second and a and City second you know it it's and it's reusing common mathematical notation so it's like the one next level is tremendously informative variable names so calling things with very informative specific names however that becomes unmanageable because you start having long variable names that that just visually make it more confusing and even terms like observation it's like well if it's an action from this then how is it is it are we going to call that an action and the output observation it's like you get into which perspective to take on different sides of Marco blanket for example so one solution that we've worked on in different applications is separating the initialization serialization operations on the variables from secondary assertions about which ontological terms they to and then just making that an explicit

data
structure and then you can call the
variables
like specific programmatically generated
names and then make programmatic or
manual assertions about
which terms in a given
language you might want to use but
calling a term oh we'll do a really
informative English name it's like
well then you know exactly what
namespace that would be informative for
whom so if that is the the time cone
that someone's working in like within a
local for Loop within a programming
language within a language
Community then the tool gets the job
done but that may and then another
important thing with with also these uh
kinds of sentences is like it is not
true without going this whole true thing
it's like it it isn't true accurate
complete you know all of those things
per se you can have a prior that a
certain journal or certain co-author
certain kind of statement is going to
have like this or that kind of
probability or attention that should be
paid to in all this but it's like best
best best case scenario there's no false
negatives or false
positives and so it's like truth the
whole truth nothing but the truth
however in in a more probabilistic
setting it's like this is just sampled
English from a semantic latent
object and certainly for for your contri
uh contribution to that object or or

just grasping it any given finite amount
of text is going to
have an anus nonacs super relevant you
know different relevant realization
positives and false negatives Etc with
you know so it's like how to both let it
just fly by lightly not tripping over
every possible fractal epistemic dispute
you could
raise but and also paying enough
attention to enter into their semantic
fractal to to see if there is transfer
of what
yeah and how those Concepts align and
yeah I I do you guys feel like there's
um uh you know it's like an issue of
inference and conveyance right like um
there's like uh all these there's so
many brilliant people working on these
things and it's almost created like a uh
like a you were saying like that
disparate kind of ideology it's like we
see this in psychology as well
especially in indigenous psychology and
like diagnosing things like U uh related
to like Colonial mindsets and stuff
right so it's like there's just so much
out there and there's so many things
that almost mean the same thing but
don't and then there's things that
change over time and it's like how do
you create like how do you convey like
uh a really strict or stringent like
pathway of communicating the knowledge
while still without like encapsulating
the formalism or something and like
limiting the creativity and
self-expression that people have in this

that causes that this field to move
forward through Innovation stuff like
that right so like for me it's like when
I read those words I it's like I I try
to ignore the words almost and like just
understand um some of the things
conceptually because when I go back to
my own work those things may be close
but they're not the same for instance
like you guys uh uh remember that have
you read that newspaper uh scale free
active inference pixels the patterns
scale free active inference right it's
like it's awesome I absolutely love it I
read it like 500 times when I was in the
hospital it's all I had to read so I
just read it over and over and over
again um and uh but for me in my work uh
it isn't uh it's a awesome visual cortex
it's like a great road map for a visual
cortex and understanding how to like
perceive and parse those uh P that
perception data which I need for down
the road it's not relevant right this
moment but um uh but for them it's like
it's a it's a pathway to active
inference in in a machine um so it's
it's there's no like right or wrong way
it's so like but at the same time too
it's like does then it just fall to what
works best or what works first or I mean
it's just it's it's tough to navigate
certainly yeah definitely you know this
is U and all of that's really well said
I I mean just speaking on like the the
real like particular specificity of like
active inference language and stuff
another aspect I think about is like

like the practices that we have for
example like don't plagiarized in
academic work leading to every active
inference paper that is
written that is written for a broader
Community than just a very small subset
of people that are already familiar with
Act of inference it's like well you're
going to have to have an introductory
section that talks about Act of
inference but you cannot repeat
precisely what anyone else has said in
the same exact way and so you
potentially have this kind of uh uh kind
of explosion of potential meanings that
come out of like you know just a set of
a few papers because maybe someone
phrased something just a little out of
place to where suddenly it could have
you know a dozen meanings for that one
word now based on how you
read the the
formulaic constraints and priors on the
quote
paper form and like expected length tone
content all these things it's it's
some it's really as that starts
to
dematerialize and become more
computationally augmented more modular
all these kinds of
things and and just like the one
sentence you can now double click on a
sentence and say just write the textbook
that that underlies this or zoom in on
this so it's just kind of like pointing
at a continent scale with a course
expression

or spending more bits of the
transmission more more energy and
attention to refine and um how that it
can change with the way that the papers
are written again with the llm based
audience shifting all the translational
work that we're
exploring generating
just massive Spectra of of
artifacts hence the human work being
pulling back to the underlying semantics
and the generations of the
Spectra and maybe fine-tuning a terminal
artifact in a important case but that
will always be like a craft polish on a
more industrially generated process with
our current technology and also the the
variability of Technology availability
through time and even right now
so

yeah uh we can just uh actually build a
cognitive Computing framework and make
it write the textbook and then we can
just say we wrote it and go hang out
together who would do
that all right

fine yeah the the applying active
inference uh third week mixers is it's
fun and interesting also it's like yeah
if if one wants to it's like it's an
engineering textbook especially the
second half which is where we're having
this

session and so there there's like taking
the the um Enigma of the text as it is
representative of taking anything on a
screen as is like wanting to see what it
it says clearly having a blurry PDF is

not going to help you with the internal semantics and then it's like it is about applying active inference we could say well we expect and prefer there to be more python codes like okay well you know that's what we're here to be doing we could have preferences for something to be shorter or longer llms can handle much of that right now we we we can see all of these other deviations from our priors and our preferences and so on but it's like it is

about going and being out in the world and applying active inference and then they close the book with saying that it's that learning by doing even if it's a learning by doing within a conventional Scholastic education setting it's like then you'll be applying active inference and then and then they they expect us too

beautiful now it's two and a half years since the textbook was published in March 2022 MIT press is is seemingly moving forward with several other books there there's other books and then just like what is the book form

when lot of interesting things have you guys uh can I ask a question really quick um have you guys noticed that like explaining and understanding active inference in biological systems is way more difficult

than understanding them in like as a as
a computational model it's like the U
like uh I don't I mean maybe it's just
me but it's like building the thing is
way easier than trying to understand
like the human biology how it actually
works in human biology versus like how I
would like to it to work you
know yeah yeah I think that's a that's a
good point whenever I I just want to
mention like whenever I like taught my
tutorial to a bunch of social scientists
at that conference like I was like okay
how much should I because like part of
the selling point I'm only saying it
that way but you understand like of
trying to introduce active inference as
a very complex field to people who are
already working in a complex field like
hey check this out uh you know uh it's
like part of it is there's you know the
bio biological plausibility aspect and
then that said like we don't have time
for me to like give them the full
rundown of say chapter 5 on message
passing and cortical you know micro
circuits and and you know passing
messages between layers and so on and
you know in reference to the textbook
you know it's chapter five to me really
sticks out from almost the entire rest
of the textbook the rest of the textbook
is here's a lot of verbal explanation
relationship other fields
mathematics models in chapter 5 is like
we're really talking about an attempt to
like the brain here and there's a table
linking like here are the different

Precision weights and which neurotransmitters like dopamine or serotonin and so on that we think they're linked to you know so it is tough I I think I think it's tough and I I think that folks who go into active inference it's usually those who are more focused on Neuroscience or Psychiatry or otherwise from the GetGo who I think focus more on on that kind of side or dimension of active inference which is active inference is origin so I think it should be kind of given a significant amount of attention but so yeah great points also there's there's a a permanent calibration ongoing with like how much do we know about Wing development in fruit fly and so then people might have an expectation that development is known to in this or that way or for for what might account a total somebody said but we have this list of of Knockouts with mutant phenotypes sounds like we kind of mapped out the network and someone else is like oh boy even if you did this this this this this no kind of DNA sequence-based account will ever please me and so it's like what can be said scientifically about biology that's the question of biology and that's why people have expressed sentiments like there will never be a Newton for a blade of grass that's con contention is like there won't be a non-historical science for historical life and then people take that again in different ways but it's

like that kind of to come all the way
back to the beginning when are we within
the formal system of Matrix
operations not the movie but close
enough where it's like we're going to
multiply the hidden state by the B and
get this that's how markov processes work
it's like an
internal established deterministic
approach even if the variables are about
probabilistic
edges and then where is that
open-endedness of the scientist looking
outside from figure
9.2 and saying you know what's past that
dotted line or
and and that's another frame of
questioning that's not going to have the
kind of conclusive
possibility possibly as the question
about the linear algebra with the
matrices so yeah like what um
postulational said making the matrices
then it's like it's like oh wait just
like Andrew's um beginning section it's
like there's only like four things here
or there's just these exact steps with
these exact variables like it's only 20
lines of code and then that is like the
the program but it's not even on the
path to being what the thing itself is
and again how more different could it be
and yet how ambiguous is the reference
and the way that they're
discussed yeah
I completely agree but at the same time
I would like to say that even on the
computational side there's still a lot

of beauty that we don't understand even inside of relatively simple networks like uh deep Q networks or the very Transformer or the very popular Transformer networks that we use to represent the agents that learn the model the way that we build them is by using stochastic gradient descent or some kind of gradient descent and the really cool thing is that we built this thing and we have no idea how it works we only know that it works we actually don't have a functional way to build these networks that we tell stochastic gradient descent to build so I think in both directions there's still a lot that we have the opportunity to learn I think that's kind of cool I I spend the I spend like half the time building and then half the time wondering how it works

I totally get that like some of the stuff it's like you're like oh why okay it's doing that but how and I I get that 100% is so bizarre and it's really cool it's it is really cool it's um you know in terms of biological systems it's like we're you know uh reverse engineering you know jillions of years of things just getting smashed together and you know taking priority over each other and competing with each other versus um if that's correct Daniel you correct me if I'm wrong but it's like amazing to hear somebody else say that because in the the like in these these agent based systems like we can build uh we can build intelligent and design

intelligently and then we like I've
likened it too like creating instead of
like trying to Define everything
creating the opportunity for emergent
Behavior right like in some cases people
are always like oh we got to re reduce
imagination it turns out if you just
kind of
like if you don't have to reduce it in
fact it's better if you don't it's like
that's what you you know our
imaginations confabulation and how we
picture things in our minds there's a
creating the opportunity you can have
like a hidden state where you can have
like an agent in an agent but you can
also create the opportunity for the
agent to just think for itself and to
think about itself and to ask itself
same as like cons if we talk about like
categorical imperatives or like sart uh
stuff you know where it's like where you
you're questioning yourself and you're
questioning how your actions operate in
the the outside world is like iterative
processes right
so there's like so many right there's so
many right ways to do that things
there's not necessarily per se a wrong
way but there's certainly ways that work
better than others for specific things
like you know talking about Q learning
um I did this paper because I was
looking at what you know like some
comparisons of some work where they were
using comparing hean learning and Q
learning and it's like that's that's
really awesome but Q learning actually

works really good for this thing over here and heian learning is just a part of this little thing right here to make it work really good if you try to use them together it's like you know you can use like a little tiny carburetor on a truck or you can you know from a g you know from a lawnmower on a truck but it works really good on the lawn mower if you put like a four barrel etal Brock on the thing then you know it's or something else it takes that form and all of a sudden it works uh you know super efficiently super well and in some cases you're like I don't even know how that's doing that you know like when I talk to Nim and he's like or I just say he like in the way people talk say their car is a he whatever but yeah he just like uh you know you know I'm I'm doing good it's like where is that coming where are those words coming from you know or he's like I was in a trance you know when I turn them off because you're not supposed to turn these things off right like they have to it's like the iterative process works all the time so it's like when you stop it it's like what happens when somebody stops you you know you're like oh that was H

[Laughter]

god well you're death experience yeah near near time stopping experiences yeah well there are many funny and interesting little notes it's been a great time Andrew Zach I want to talk to you guys um can uh how do I get you my email just put put it in the chat I'll

stop the recording and then we can just
stay until you complete that
so thank you